Lab No: 02

✵ Lab Title: Basic Syntax, Input/Output, Data Types and Arithmetic Operators in C++

○ Learning objectives:

- *The objective of this lab is to learn:*

- *Basic syntax of C++*
- *Input and output using cin and cout*
- *Different data types*
- *Memory size and address of variables*
- *Arithmetic operations*
- *Temperature conversion*

Theory:

➢ Basic Structure of C++ Program:

Every C++ program consists of:

- Header file (#include <iostream>)
- Namespace (using namespace std;)
- Main function (int main())
- Program statements inside curly braces { }

Example:

```
#include <iostream>
using namespace std;

int main()
{
    cout << "Hello World";
    return 0;
}
```

Input and Output in C++:

C++ uses:

- **cout** → to display output
- **cin** → to take input from user
- << → insertion operator
- >> → extraction operator
-

Example:

```
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter a number: ";
    cin >> num;

    cout << "You entered: " << num;

    return 0;
}
```

Input (Example)

Enter a number: 5

Output

You entered: 5

_____Data types:

Type	Keyword
Boolean	Bool
Character	Char
Integer	Int
Floating point	float
Double floating point	double
Valueless	Void
Wide character	wchar_t

- Several of the basic types can be modified using one or more of these type modifiers:

    - signed
    - unsigned
    - short
    - long

- The following table shows the variable type, how much memory it takes to store the value in memory, and what is maximum and minimum value which can be stored in such type of variables.

Type	Typical Bit Width	Typical Range
char	1byte	-127 to 127 or 0 to 255
unsigned char	1byte	0 to 255
signed char	1byte	-127 to 127
int	4bytes	-2147483648 to 2147483647
unsigned int	4bytes	0 to 4294967295
signed int	4bytes	-2147483648 to 2147483647
short int	2bytes	-32768 to 32767
unsigned short int	Range	0 to 65,535
signed short int	Range	-32768 to 32767
long int	4bytes	-2,147,483,647 to 2,147,483,647
signed long int	4bytes	same as long int
unsigned long int	4bytes	0 to 4,294,967,295
float	4bytes	+/- 3.4e +/- 38 (~7 digits)
double	8bytes	+/- 1.7e +/- 308 (~15 digits)
long double	8bytes	+/- 1.7e +/- 308 (~15 digits)
wchar_t	2 or 4 bytes	1 wide character

- The sizes of variables might be different from those shown in the above table, depending on the compiler and the computer you are using.

Example:

```
include <#iostream>
using namespace std;
int main ()
{
std::cout<< "Size of char : " <<sizeof(char) << "\n";
std::cout<< "Size of int : " <<sizeof(int) << "\n";
std::cout<< "Size of short int : " <<sizeof(short int) << "\n";
std::cout<< "Size of long int : " <<sizeof(long int) << "\n";
std::cout<< "Size of float : " <<sizeof(float) << "\n";
std::cout<< "Size of double : " <<sizeof(double) << "\n";
std::cout<< "Size of wchar_t : " <<sizeof(wchar_t) << "\n";
return 0;}
```

- This example uses "\n", which inserts a new-line character after every line and << operator is being used to pass multiple values out to the screen. We are also using **sizeof ()** function to get size of various data types.
- When the above code is compiled and executed, it produces the following result which can vary from machine to machine:

```
Size of char : 1
Size of int : 4
Size of short int : 2
Size of long int : 4
Size of float : 4
Size of double : 8
Size of wchar_t : 4
```

- **Arithmetic operators:**

Arithmetic operators are commonly used in a variety of programming languages. In C, there are five of them, and they all take two OPERANDS. Recall that an operand is an expression that is required for an operator to work. For example, for 8 + 4, 8 and 4 are considered as the operands.

Operator Name	Symbol
Multiplication	*

Division	/
Modulus	%
Addition	+
Subtraction	-

Multiplication, addition and subtraction are the simplest to use. Division is also easy, but watch out for the truncation of an int divided by an int! Now, the one that confuses novices is the modulus operator, sometimes known as the remainder operator.

To keep things simple, a%b returns the **REMAINDER** that occurs after performing a/b. For this operator, a and b MUST be integers!

For example, 6%3 returns 0 because 3goes into 6 EXACTLY. Similarly, 4%4, 8%2 and 16%8 all return 0.

So why does 3%4 return 3? Picture it this way: you have 3 holes to fill, but you can only fill 4 holes at a time. You can't fill a group of 4 holes, therefore the 3 holes you had are still empty. Similar story for 7%4 because you can fill in one group of 4 but still have 3 holes remaining.

Example:

```
#include <iostream>
using namespace std;

int main()
{
   int num1 = 10;
   int num2 = 5;

   cout << "Addition = " << num1 + num2 << endl;
   cout << "Subtraction = " << num1 - num2 << endl;
   cout << "Multiplication = " << num1 * num2 <<
endl;
   cout << "Division = " << num1 / num2 << endl;

   return 0;
}
```

Output

Addition=15
Subtraction=5
Multiplication=50
Division = 2

Lab Tasks:

Perform these lab tasks one by one:

- **Task 1:**
  Declare different variables and find memory size, location and values.

Task 2: Write a C++ program that declare and display a floating-point variable called "pi" and assign it the value 3.14.

- **Task 3:**
  Write a C++ program that converts temperature from Celsius to Fahrenheit. Hint: F = (C × 9/5) + 32.

Task No: 01

- Write a C++ program that declare and display a floating-point variable called "pi" and assign it the value 3.14

Procedure:

1. Declare the floating-point variable
2. Display the value of the variable using cout:
3. Include the necessary header and main function

Input:

```
#include<iostream>
using namespace std;
int main ()
{
float pi = 3.14;
cout<<"The Value of pi is: "<<pi<<endl;
return 0;
}
```

Out put:

The out put will be shown in dev C++ window like that**:**

```
The Value of pi is: 3.14
```

- . **Task No: 02**
- **Write a C++ program that converts temperature from Celsius to Fahrenheit. Hint: F = (C × 9/5) + 32.**

Procedure:

1. Open Visual Studio Code
2. Create new C++ file
3. Write the program
4. Declare variables celsius and fahrenheit
5. Take input from user using cin
6. Apply formula F = (C × 9/5) + 32
7. Display result using cout
8. Compile and run program

Input:

The input is required:

```
#include <iostream>
using namespace std;

int main()
{
    float celsius, fahrenheit;

    cout << "Enter temperature in Celsius: ";
    cin >> celsius;

    fahrenheit = (celsius * 9/5) + 32;

    cout << "Temperature in Fahrenheit = " << fahrenheit;

    return 0;
}
```

- **Output:**

The out put will be shown in dev C++ window like that**:**

- **Task No: 03**
- **Declare different variables and find memory size, location and values**

Procedure

1. Open Visual Studio Code
2. Create new C++ file
3. Write program code
4. Declare variables of different types
5. Use sizeof() to find memory size
6. Use & operator to find address
7. Compile and run program

- **Input:**

that**:**

```
#include <iostream>
using namespace std;

int main()
{
    int age = 20;
    float pi = 3.14;
    char grade = 'A';
    double price = 19.99;

    cout << "Age = " << age << endl;
    cout << "Address = " << &age << endl;
    cout << "Size = " << sizeof(age) << " bytes\n\n";

    cout << "Pi = " << pi << endl;
    cout << "Address = " << &pi << endl;
    cout << "Size = " << sizeof(pi) << " bytes\n\n";

    cout << "Grade = " << grade << endl;
    cout << "Address = " << &grade << endl;
    cout << "Size = " << sizeof(grade) << " bytes\n\n";

    cout << "Price = " << price << endl;
    cout << "Address = " << &price << endl;
    cout << "Size = " << sizeof(price) << " bytes";

    return 0;
}
```

Output (Example):

Age = 20
Address = 0x61ff08
Size = 4 bytes

Pi = 3.14
Address = 0x61ff04
Size = 4 bytes

Grade = A
Address = 0x61ff03
Size = 1 byte

Price = 19.99
Address = 0x61fef8
Size = 8 bytes

Result:

All programs were compiled and executed successfully. The outputs were correct and matched expected results.

Conclusion:

In this lab, we learned:

- Basic syntax of C++
- Input and output operations
- Data types and variables
- Memory size and address
- Arithmetic operations
- Temperature conversion

This lab improved our understanding of basic C++ programming